Quantum State Stabilisation

The *PROtoQol*™

Theoretical formulation

- Open system dynamics of the form

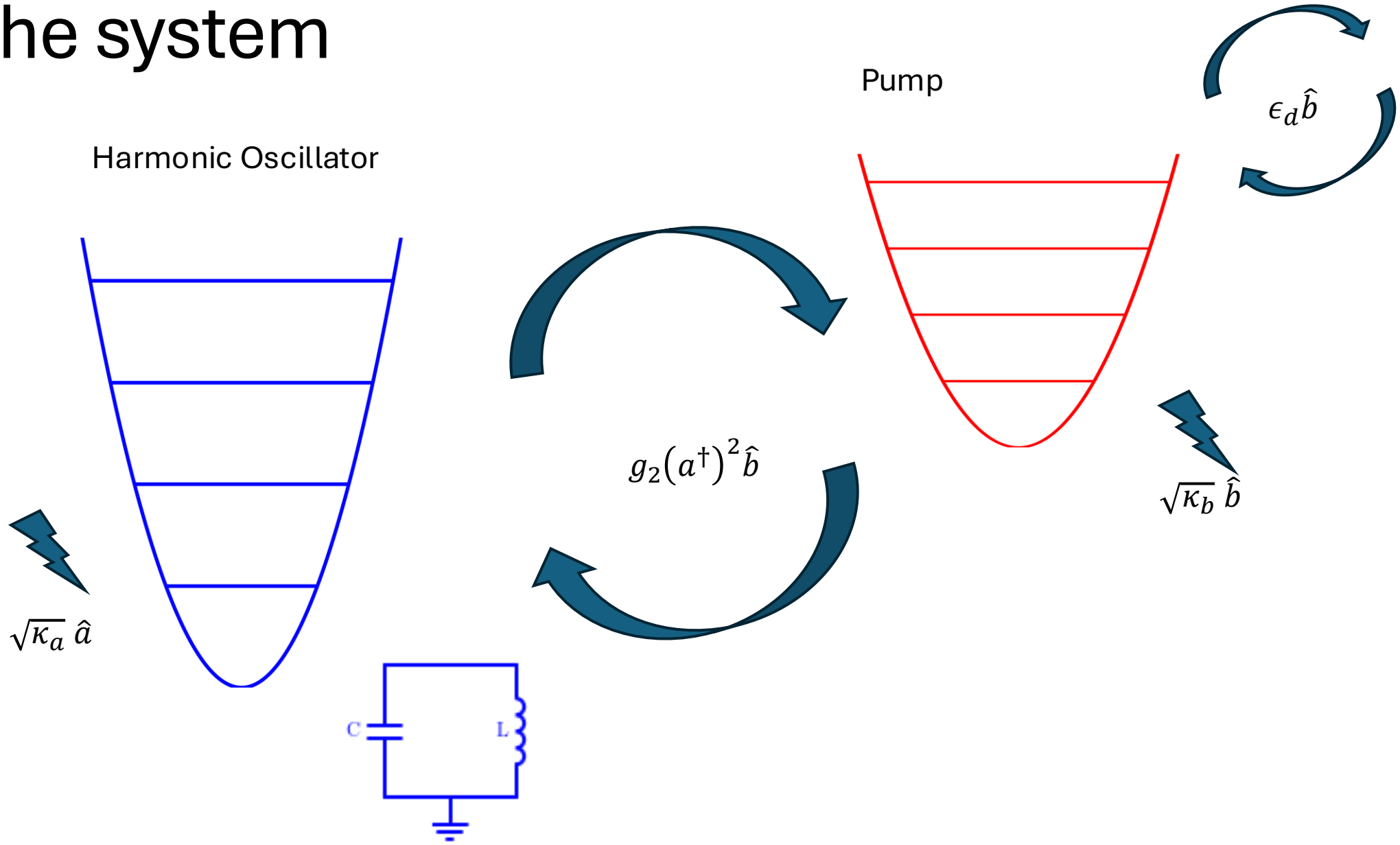
$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho} = \mathcal{L}[\hat{\rho}] = -i[\hat{H}, \hat{\rho}] + \sum_{i \geq 0} \gamma_i \left[\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \frac{1}{2}\{\hat{L}_i \hat{L}_i^\dagger, \hat{\rho}\} \right]$$

- Dissipative driven system

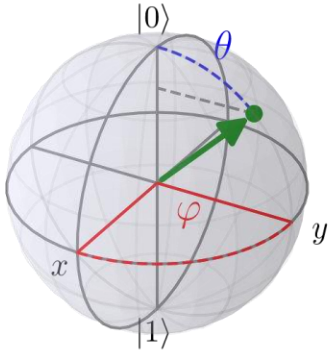
$$\hat{H} = g_2(\hat{a}^\dagger)^2 \hat{b} + g_2(\hat{a})^2 \hat{b}^\dagger - \varepsilon_d \hat{b}^\dagger - \varepsilon_d \hat{b}$$

$$\hat{L}_a = \sqrt{\kappa_a}\hat{a} \ , \ \hat{L}_b = \sqrt{\kappa_b}\hat{b}$$

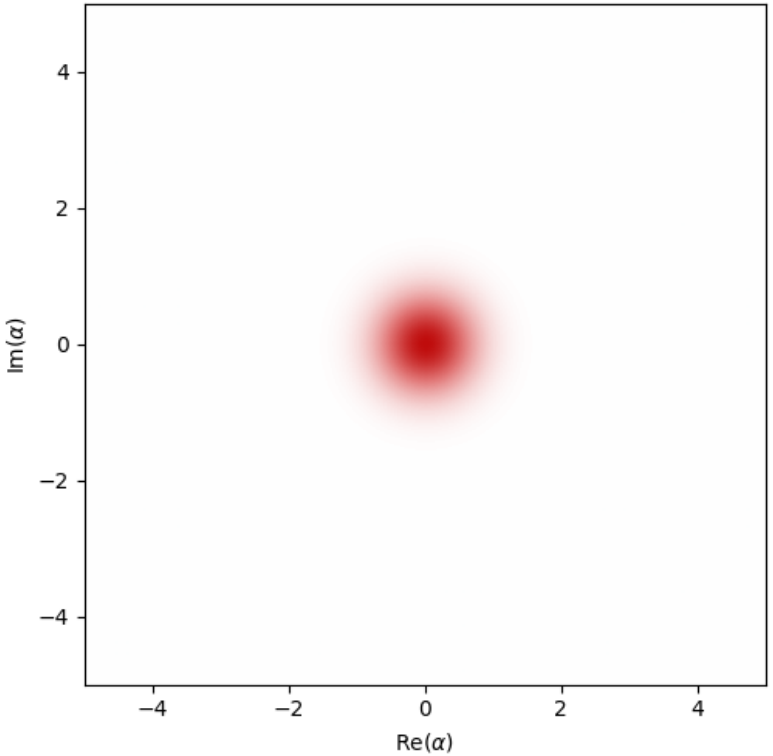
The system


Pump
$\epsilon_d \hat{b}$
Harmonic Oscillator
$g_2 (a^\dagger)^2 \hat{b}$
$\sqrt{\kappa_b}\ \hat{b}$
$\sqrt{\kappa_a}\ \hat{a}$
C
L

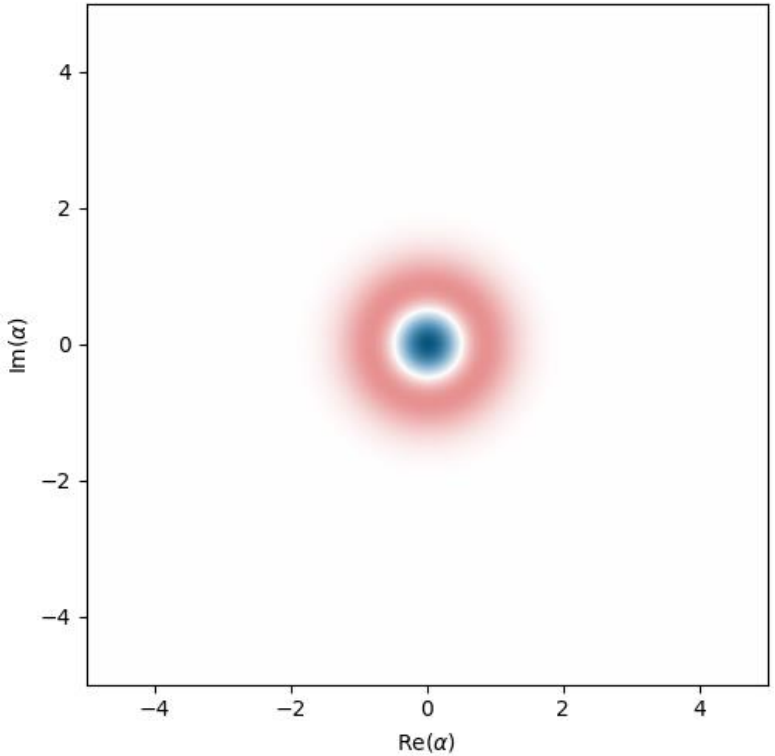
Growing orthogonal states



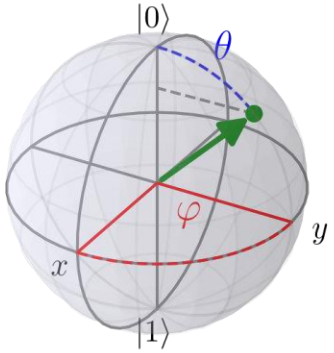
Logical $|+\rangle$



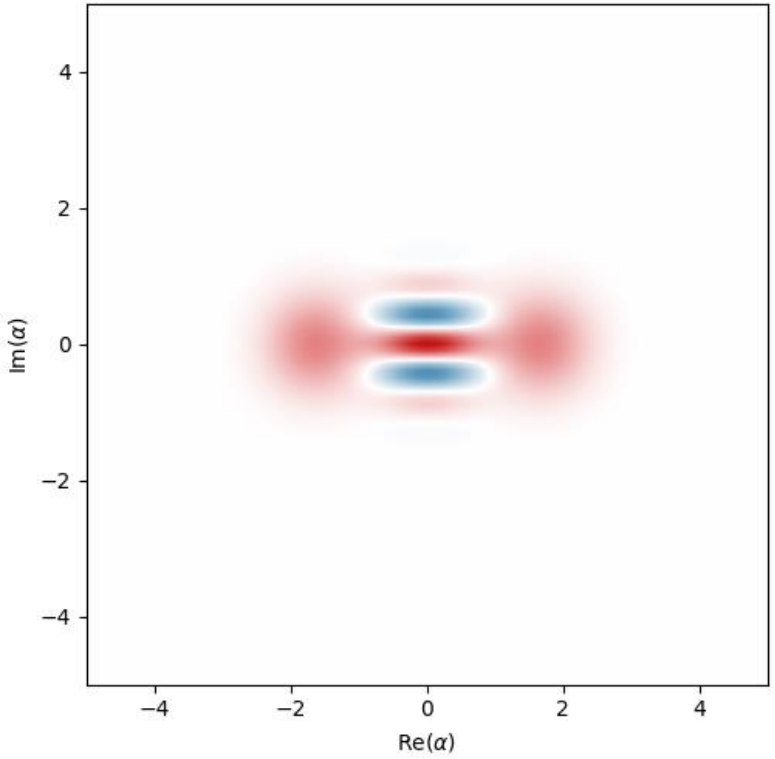
Logical $|-\rangle$



Applying gates



Pauli Z evolution



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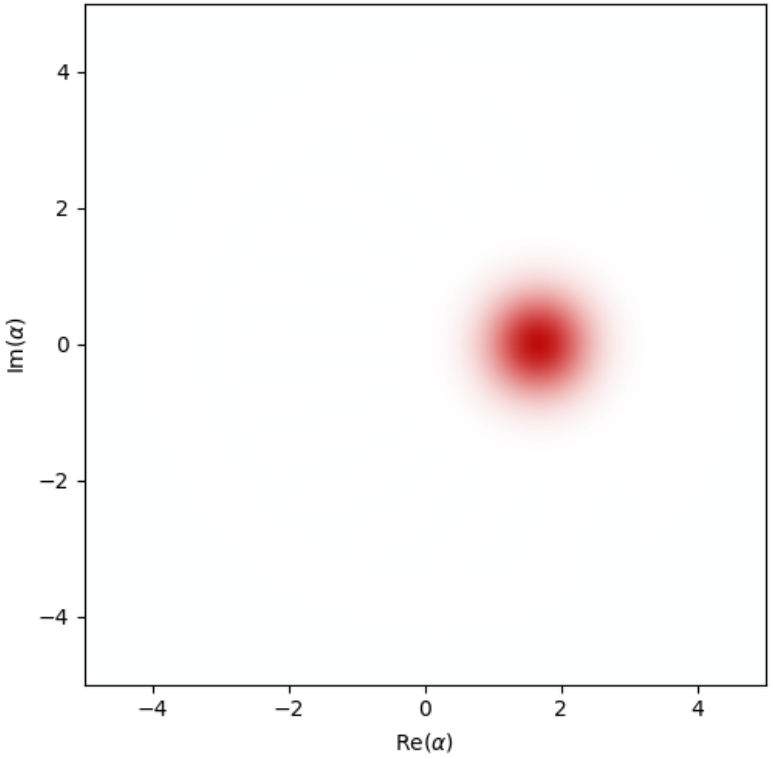


The challenge vs The solution

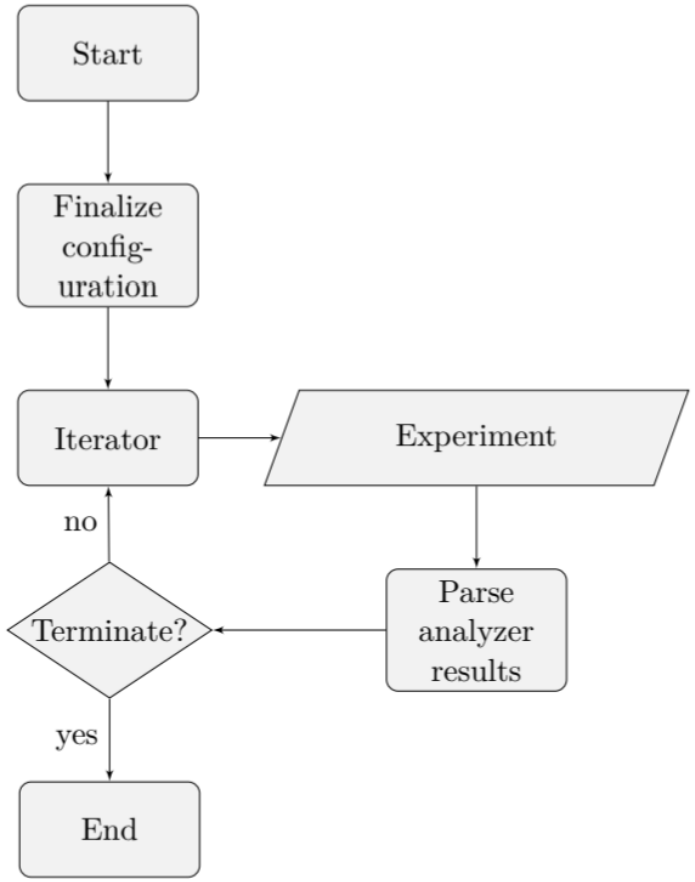
The challenge:

Traditional procedural calibration approaches struggle with hardware drift and there exists a considerable gap between their effectives in controlled simulation versus what is actually realised in practice. Dissipative cat qubits require complex tuning of the two-mode coupling the storage system pumping in order to maintain high-noise bias and logical coherence.

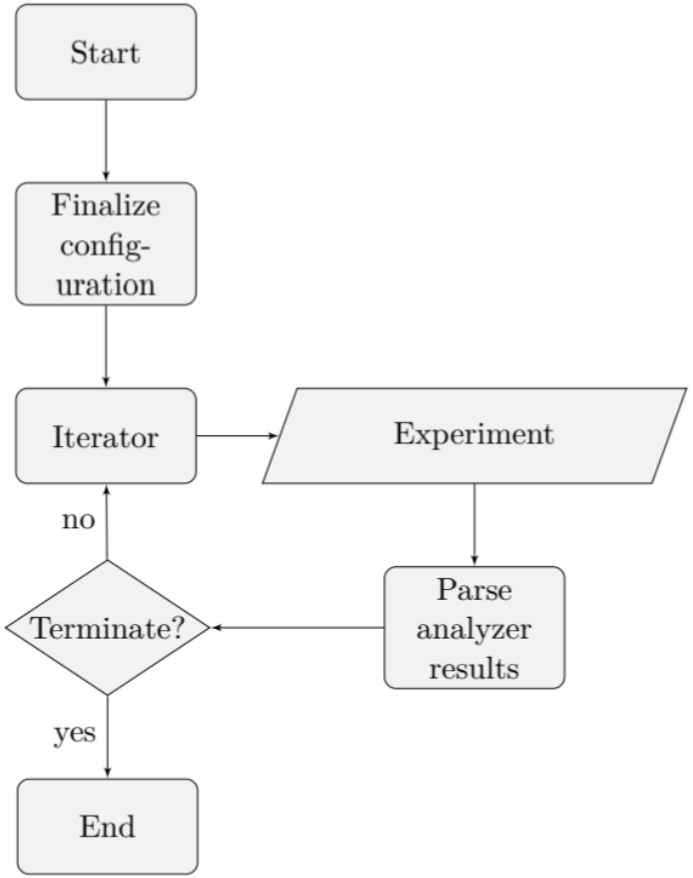
The ***PROtoQol***™ solution:

We explored the application of an autonomous reinforcement learning agent that performs an online calibration procedure on the system. By utilising partial policy optimisation, it tracks the Hamiltonian drift in real-time and maintains modifies the control parameters so as to retain the desired target bias whilst maximising the pertinent coherence times.

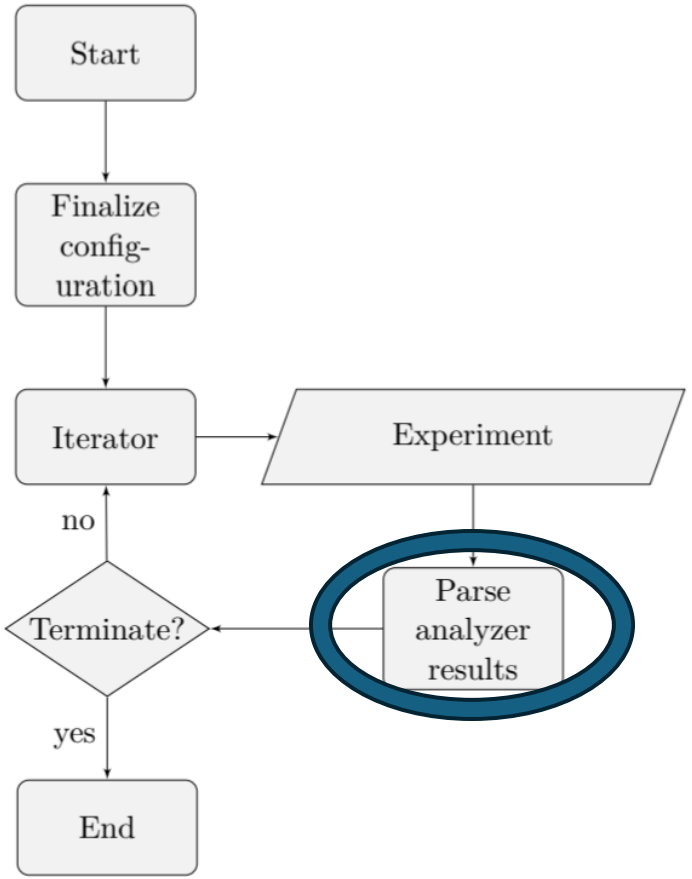
PROtoQol™ feedback & learning loop



PROtoQol™ feedback & learning loop



PROtoQol™ feedback & learning loop



PPO

- Neural network
- Optimizer actor-critic model
- With reward function:

$$R = \log T_x + \log T_z + d\left(\frac{T_x}{T_z}, \eta\right)$$

- With Loss function:

$$L = T(R) + H$$

Actor-Critic Architecture

The Actor

This consists of a policy network that predicts the optimal adjustments for the two-mode coupling and the buffer driving term based on the current state of the system.

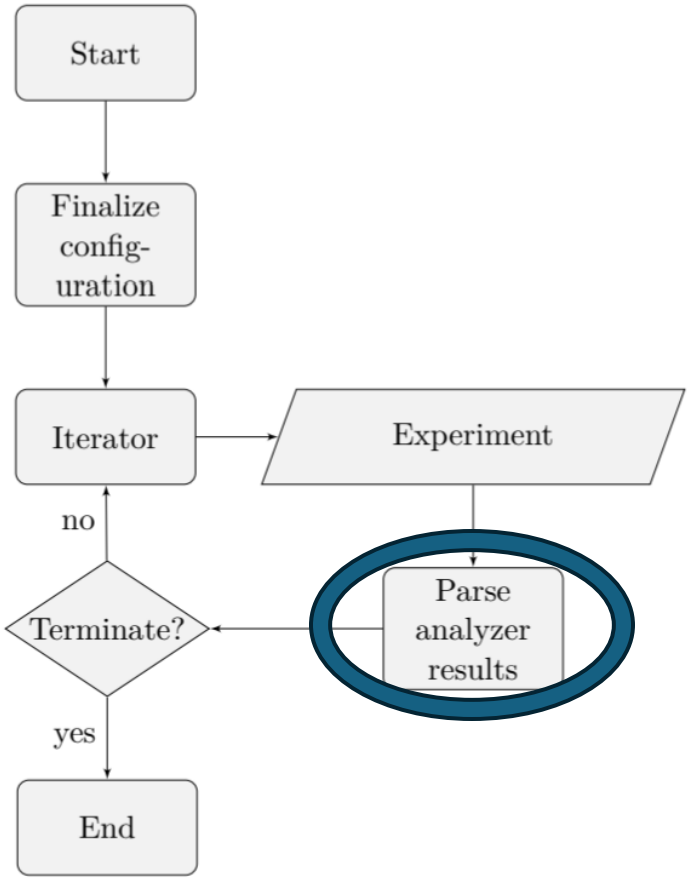
The Environment

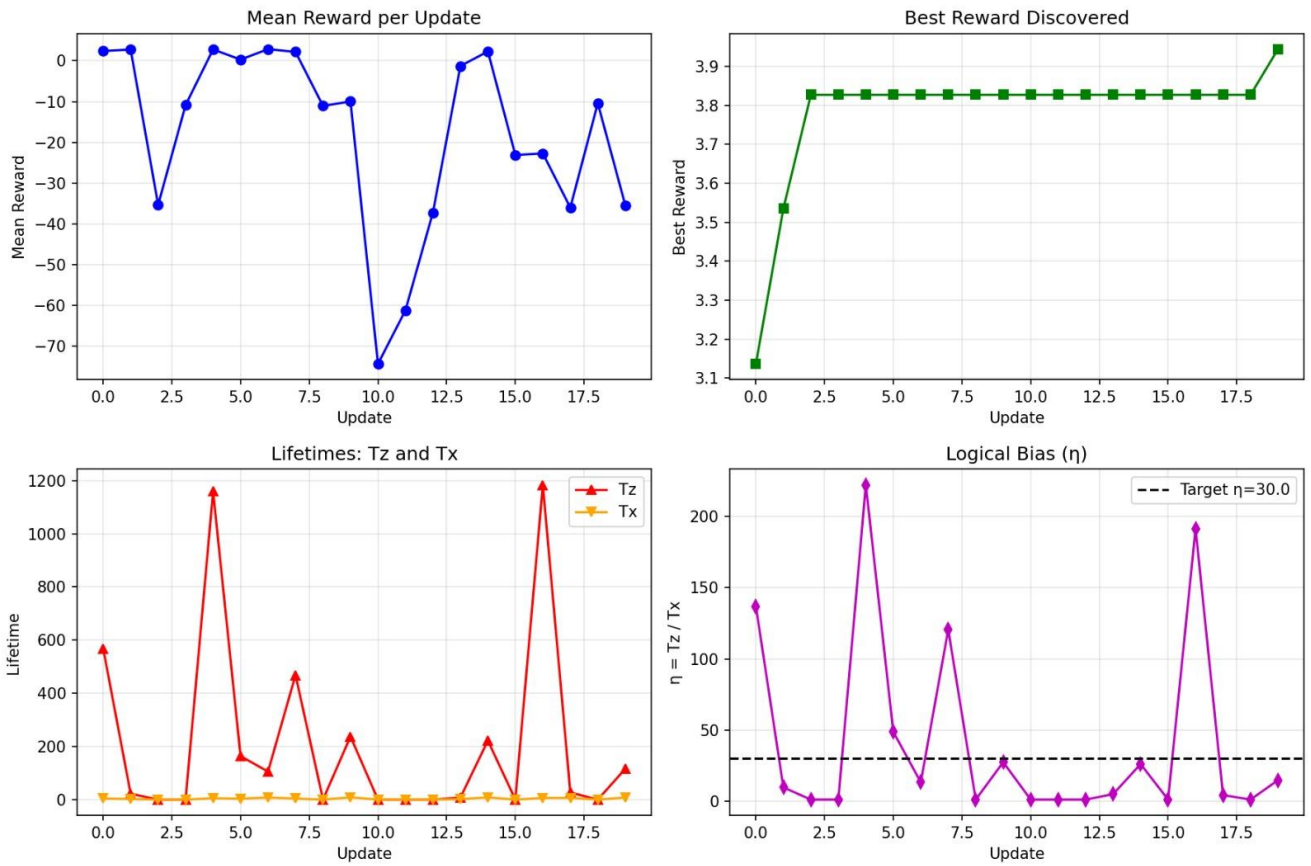
The composite quantum system, composed of the storage mode (a) and the buffer mode (b), which is subject to time-resolved dynamics under non-stationary stochastic drift.

Reward Shaping

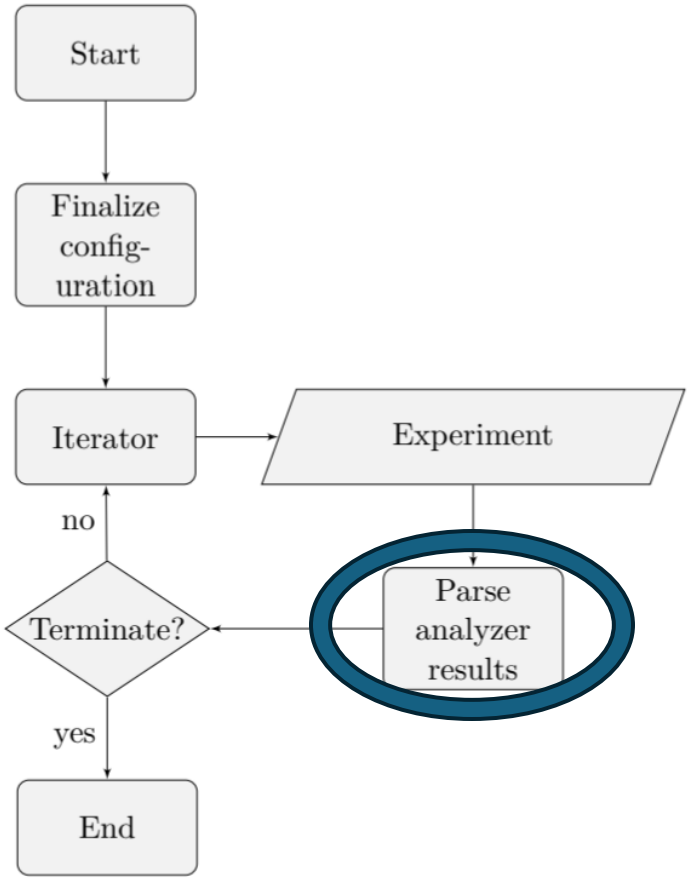
In order to achieve the desired optimisation objectives, a composite reward function is employed in order to balance between the geometric mean lifetime of bit/phase flips, bias accuracy, and physical safeguards against photon overflow.

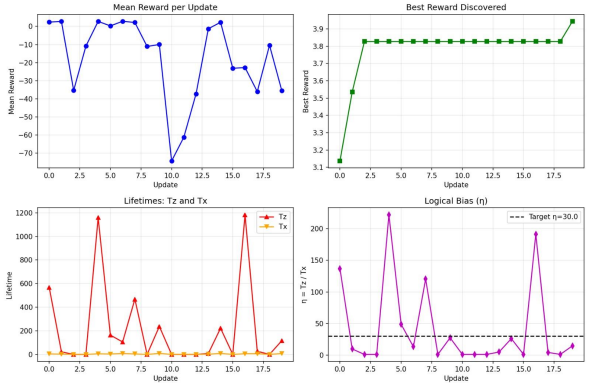
PROtoQol™ feedback & learning loop



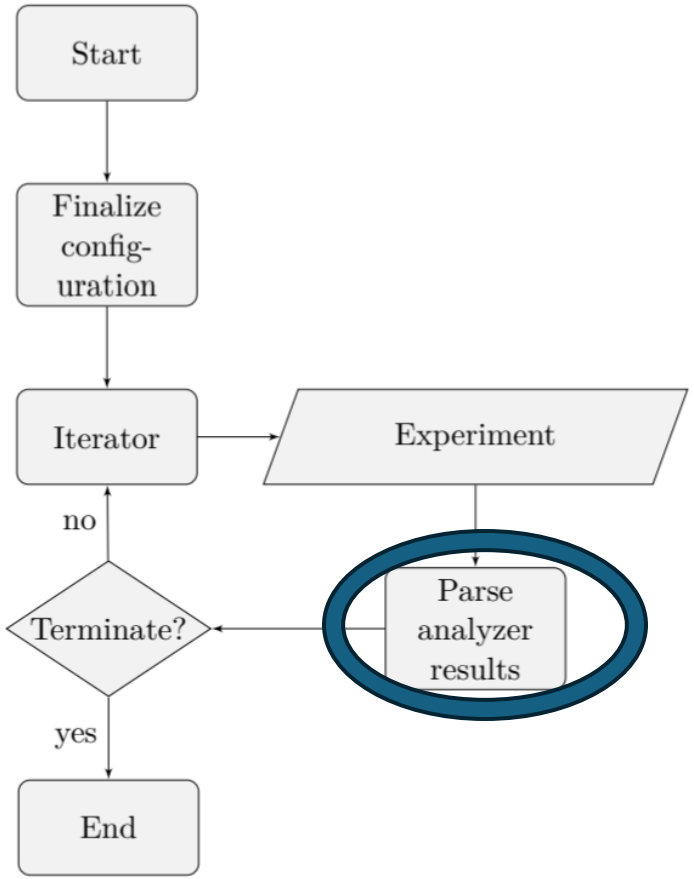


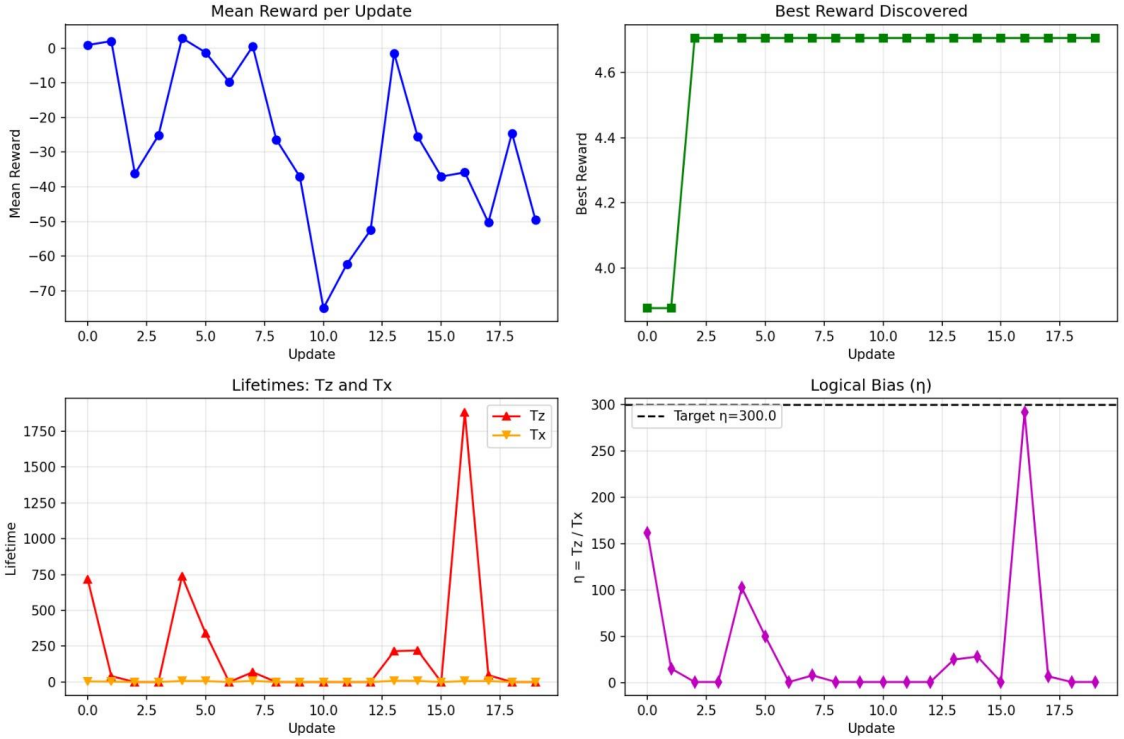
***PROtoQol*™**
feedback & learning loop





PROtoQol™
feedback & learning loop





PROtoQol™ Pseudo-code

```
• Initialize PPO actor-critic network, optimizer
• For each PPO update round:
•     For each rollout step:
•         1. Actor samples a knob vector: x = [g2, eps_d]
•         2. Build full two-mode Hamiltonian H(g2, eps_d)
•         3. Find the stabilized cat manifold directly
•            - Achieved by simulating the full Lindblad dynamics and tracing out the buffer mode
•            - Locate the dominant coherent state lobe.
•         4. Estimate the T_Z/T_X lifetimes
•            - Initialise the system in logical Z/X-basis state
•            - Evolve under full Lindblad dynamics with stochastic jumps in between
•            - Measure the time dependence of the logical Z/X observable, including noise effects
•            - Fit noisy decay curve, extract the lifetime, and train a model to predict its value
•         6. Compute the bias ratio eta = T_Z / T_X
•         7. Compute reward (maximize T_Z and T_X, penalize eta deviations and unphysical states)
```